Dimensions³

Multi-satellite flood evidence for the April 2024 UAE storm, in a portal anyone can use

TEAM
Dimensions³

THEME
Climate-induced disasters

COUNTRY
United Arab Emirates

THE PROBLEM THE PORTAL SOLVES

There is no single platform that connects the datasets, builds what the user needs, and explains it to everyone

01

Scattered data

Satellite images and data sit on separate platforms and sources, each in its own format. No one place links them.

02

No packet built for the user

Users must search image by image. No system reads the elements in the data and groups the right pictures into one package based on the user's needs.

03

Analysis only experts can do

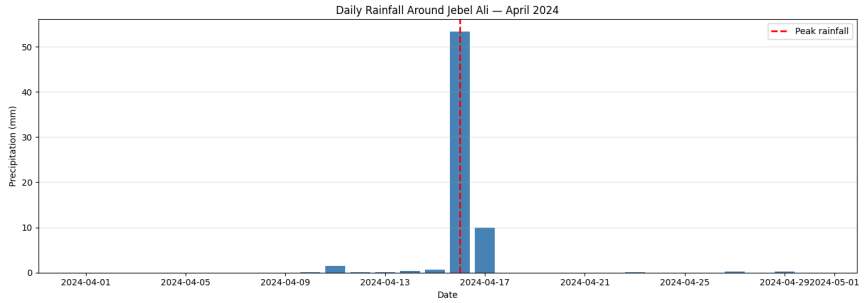
Processing is complicated and results are technical. Most students and decision-makers need easy analysis and readable reports.

Climate change has many drivers, so the analysis must consider many of them together

Rainfall · Surface water · Heat · Vegetation · Soil moisture · Dust · NO_2 · SO_2 · CO · Methane & CO_2 · Land use · Coastline

THE POC PROBLEM

Flood intelligence arrives too slowly after extreme rainfall



Jebel Ali, Dubai: 53.39 mm peak rainfall on 16 April 2024 (NASA POWER).

Separate sources

Radar, optical imagery, rainfall and reports sit on separate platforms; teams combine them by hand.

Specialist processing

Raw imagery needs remote-sensing expertise before it supports a decision.

Who is affected

Residents and commuters, port and free-zone operators, and the emergency and municipal teams whose inspection and response are delayed.

BUSINESS USE CASE & END USER

A flood evidence service for authorities, delivered through a simple portal

End user

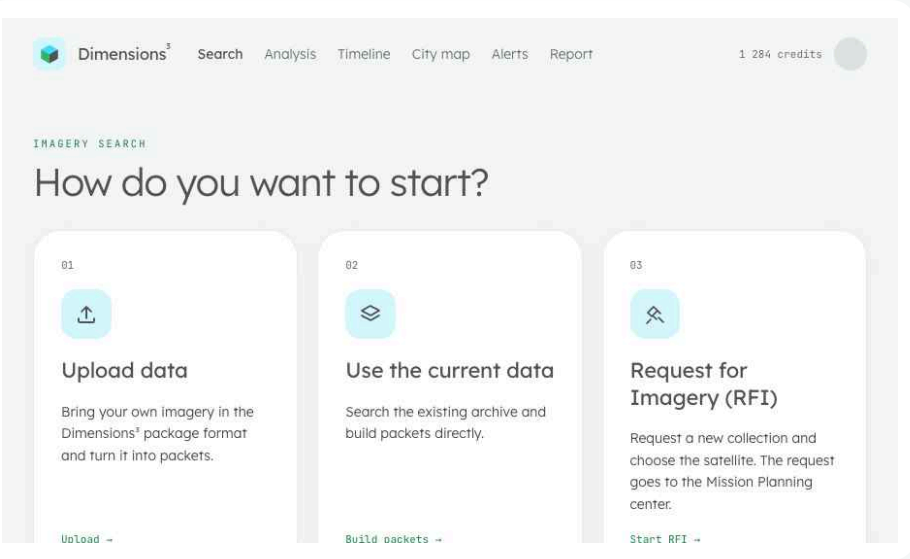
Emergency-management and municipal staff. Not remote-sensing experts: they choose an area and dates, and get maps, figures and a confidence score.

Use case

After an event: where did water stand, how sure are we, which sites were hit. Results feed response, insurance claims and drainage planning.

Model

Subscription for government entities; per-report pricing for ports, insurers and developers.



Portal: the starting screen

Upload data, use the current archive, or request new imagery (RFI).

DATA

Five sources, all open, fused over one study area

SOURCE	DATES	DETAIL	ROLE
Sentinel-1A SAR	12 → 24 Apr 2024	IW GRD, VV + VH, ascending	All-weather radar change
Sentinel-2 L2A	12 → 22 Apr 2024	Tile 40RBN, 20 m, B02–B11	Optical water change (MNDWI)
Landsat 9 L2SP	3 → 19 Apr 2024	Path/row 160/043, 30 m	Independent optical evidence
Rainfall	1–30 Apr 2024	NASA POWER, 24.98°N 55.05°E	Event support: peak 53.39 mm
Reports	Apr 2024	6 official/news + 3 public	Regional corroboration

Hyperspectral data

Not used in the current flood result. Future use: surface moisture, material signatures and water quality where suitable hyperspectral imagery exists (e.g. S813). The portal already accepts hyperspectral cubes (ENVI, HDF5, GeoTIFF). Sources: Copernicus Data Space, Microsoft Planetary Computer, USGS Landsat, NASA POWER.

APPROACH

The workflow we built and ran in Python (Colab)

01
Define area & event
Jebel Ali box, 1–30 Apr, event 15–17 Apr

→

02
Collect scenes
STAC search: Sentinel-1, Sentinel-2, Landsat 9

→

03
Water indices
MNDWI before/after for optical sensors

04
Radar change
Sentinel-1 VV/VH backscatter drop = water

→

05
Add context
Rainfall and 9 report records

→

06
Fuse
Common 374 × 341 grid; keep pixels backed by ≥ 2 sensors

→

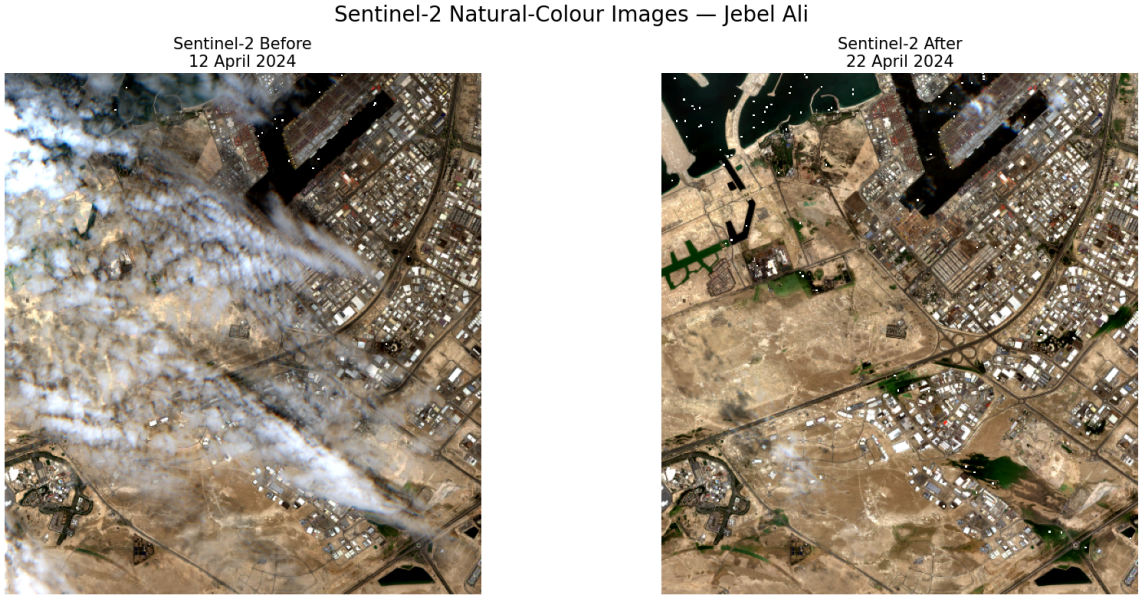
07
Refine
Remove permanent water, coastline and noise

Output
Combined evidence score = 50% spatial support + 30% rainfall support + 20% supporting evidence. Labels: low < 0.45, medium 0.45–0.70, high ≥ 0.70.

Into the portal
Results become a packet: figures, numbers and a plain-language summary.

EXAMPLE OUTPUT · OPTICAL

Sentinel-2: before and after the storm

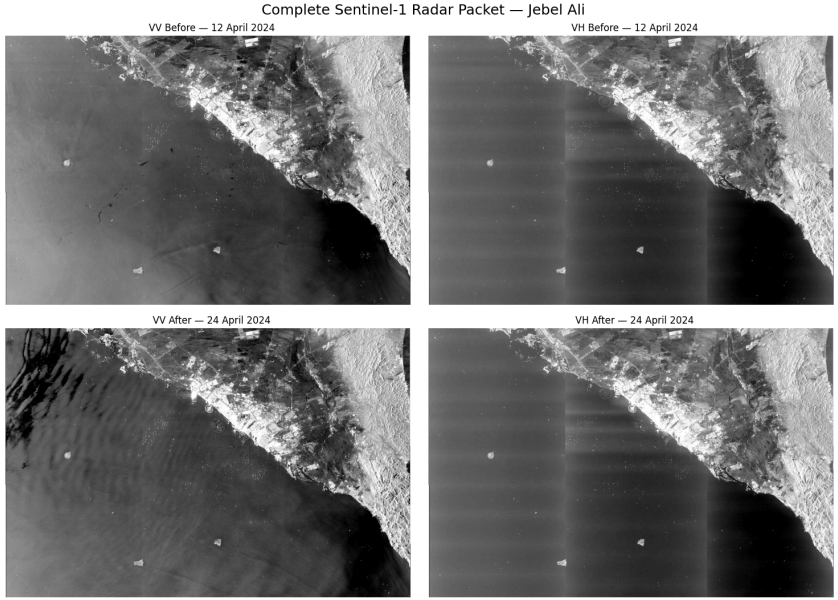


Sentinel-2 natural colour · 12 Apr vs 22 Apr 2024

Smoke and cloud cover the area before; the after scene gives a clear view of the port and industrial zone.

EXAMPLE OUTPUT · RADAR

Sentinel-1 radar sees the ground through cloud

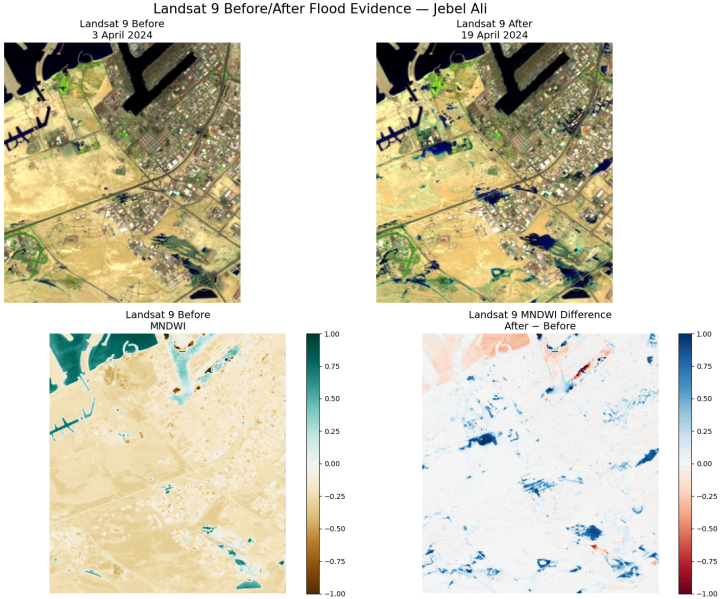


Sentinel-1 VV and VH · 12 Apr vs 24 Apr 2024

Calibrated backscatter in dB. Standing water reflects the radar away and appears dark.

EXAMPLE OUTPUT · WATER INDEX

Landsat 9: new surface water inland

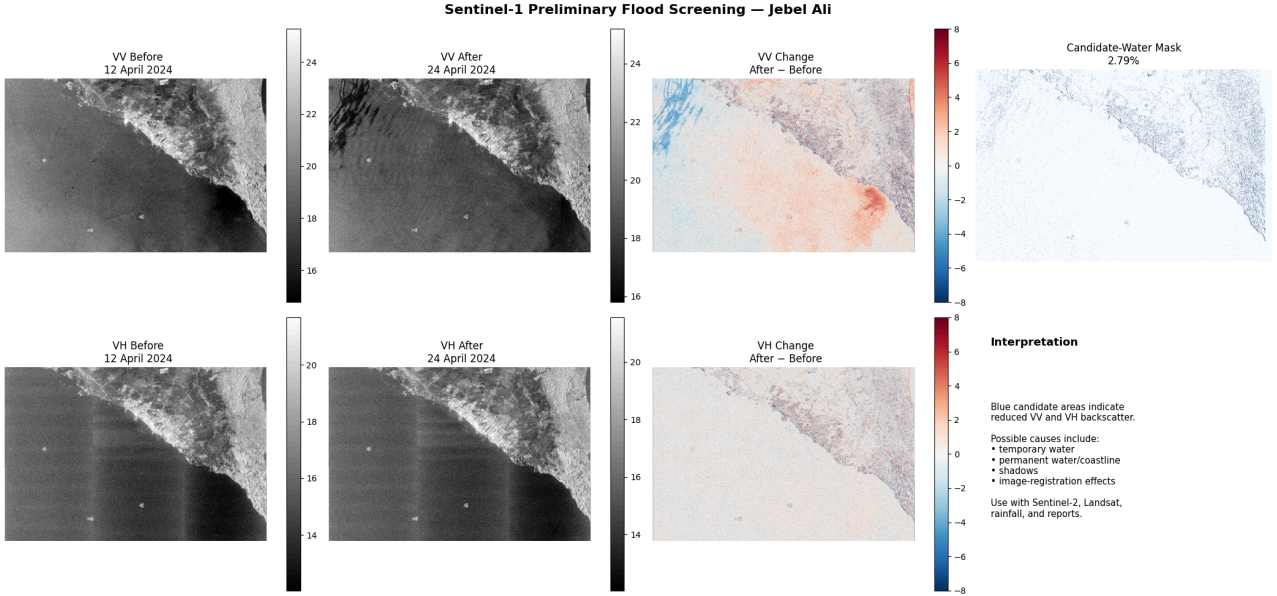


Landsat 9 colour and MNDWI change · 3 Apr vs 19 Apr 2024

Blue patches in the change map are new water on 19 April, three days after the rain.

EXAMPLE OUTPUT · RADAR SCREENING

Sentinel-1 flood screening: 2.7% candidate water

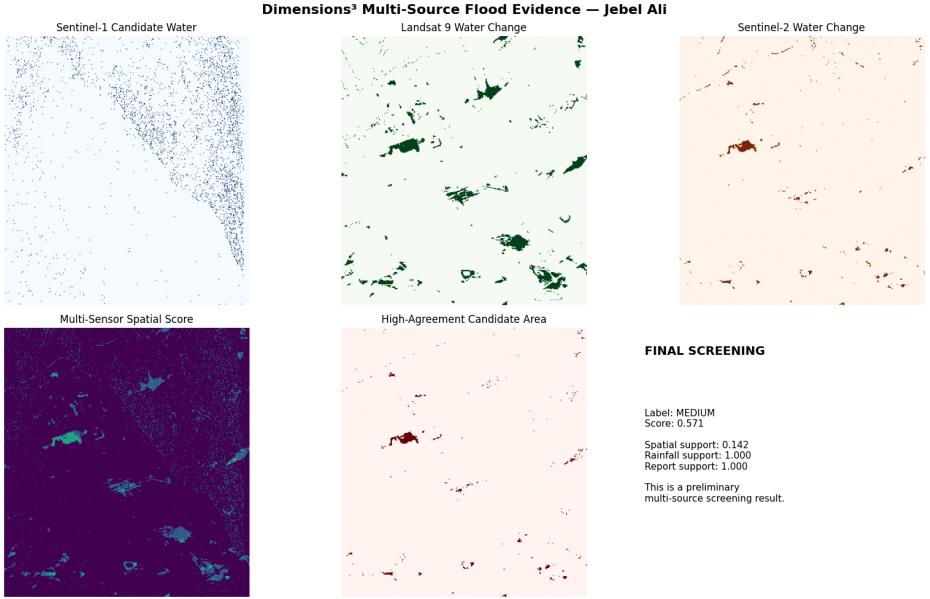


Sentinel-1 change maps and candidate-water mask

25,396 of 909,030 valid pixels flagged as candidate water.

EXAMPLE OUTPUT · FUSION

Three satellites combined into one evidence map

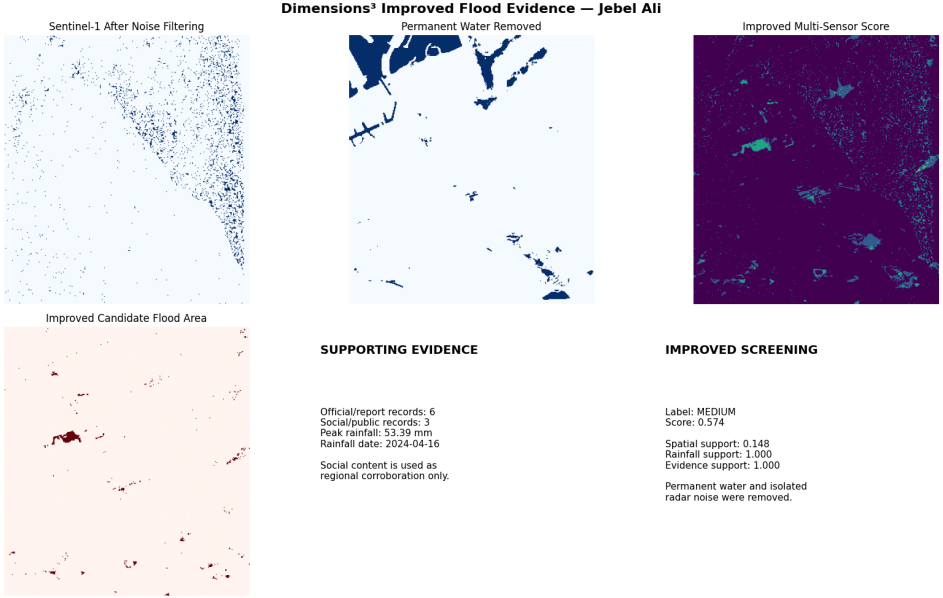


Multi-source fusion: Sentinel-1, Landsat 9, Sentinel-2

Each sensor's mask, the weighted score and the high-agreement candidate area.

EXAMPLE OUTPUT · FINAL SCREENING

Final result: 0.74% high-agreement flood area, score 0.574

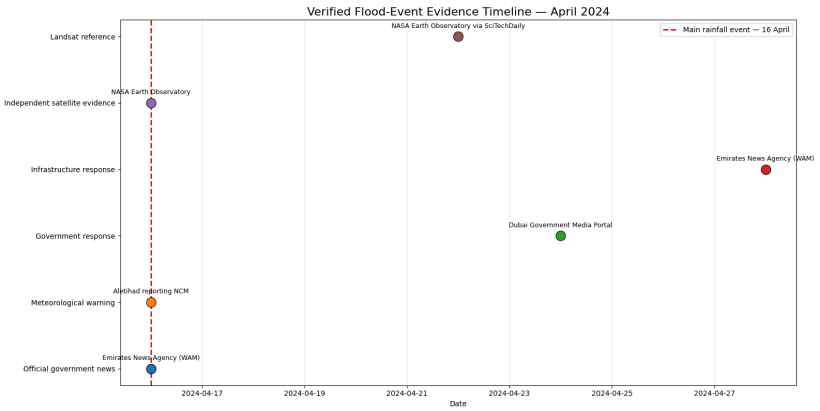


Improved screening with permanent water removed

9,207 sea and port-basin pixels removed; 873 pixels supported by at least two sensors. MEDIUM screening label.

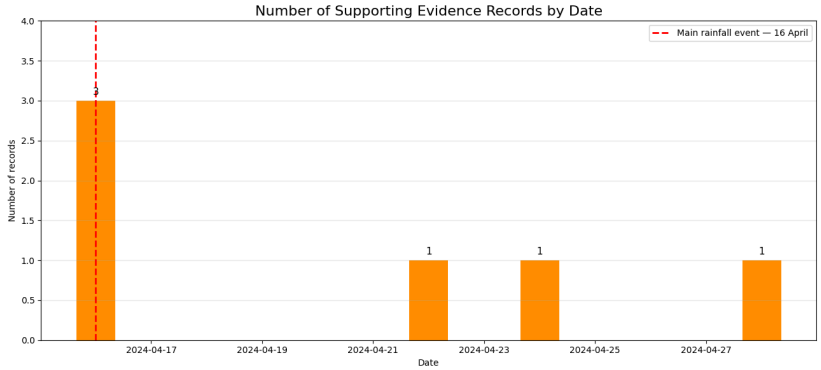
EXAMPLE OUTPUT · EVIDENCE

Independent evidence lines up with the satellite signal



Evidence timeline

Official, news and public records against the 16 April rainfall peak.

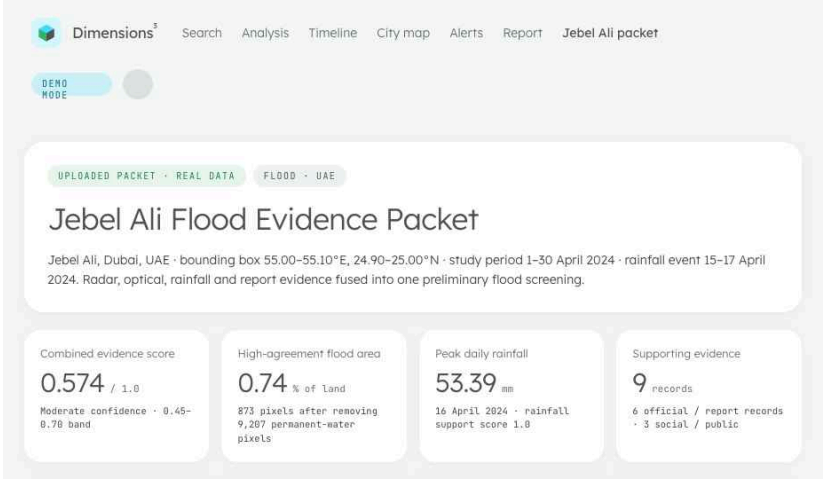


Records by date

Most reports on 16–17 April, follow-ups until 27 April.

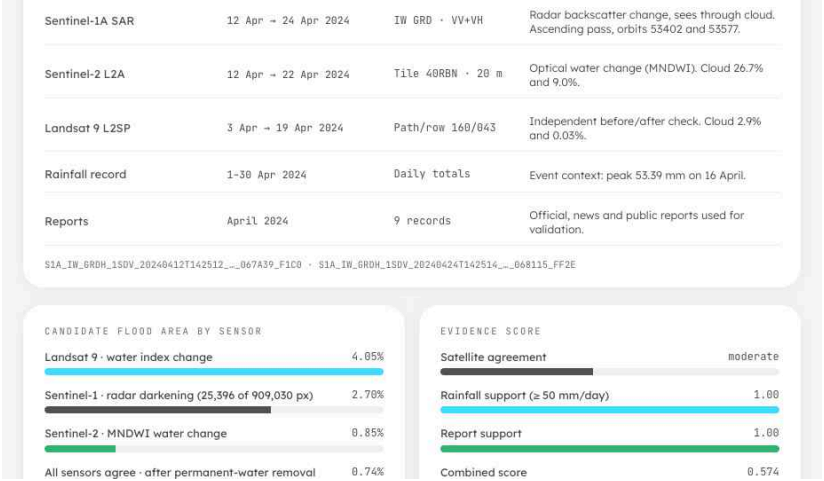
EXAMPLE OUTPUT · PORTAL

The same results in the portal, for non-experts



Packet page: headline figures

Score, flood area, rainfall and evidence in plain words.



Packet page: evidence by sensor

What each satellite found, and how the score is built.

UI / UX · THE PORTAL IDEA

One guided journey: from a question to an answer, without touching raw data

01
Ask
Upload data, use the archive, or request imagery

→

02
Packet
AI groups matching images; user ticks what to keep

→

03
Analyse
Maps, emissions and disaster patterns

→

04
Compare
Timeline with AI-filled gaps and delta

→

05
Locate
Affected streets on the city map

→

06
Act
Alerts to authorities and one full report

Plain language
Results as keywords, icons and short sentences, never raw pixel values.

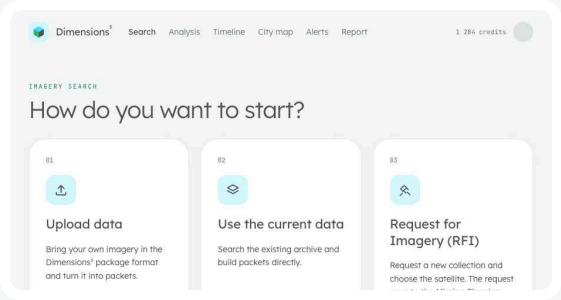
Guided inputs
Dates, seasons and surfaces picked from menus; no code or GIS tools.

Visible confidence
Every result carries a score and a low / moderate / high label.

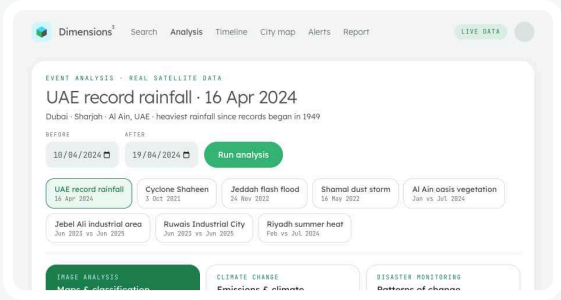
Calm, light interface
Light background, one green accent, clear hierarchy for long sessions.

UI / UX · SIX SCREENS

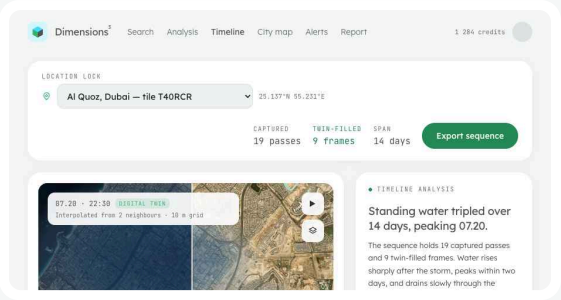
Every module of the portal, designed and clickable



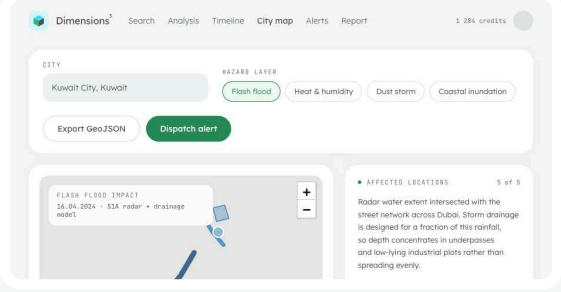
Search
Three ways to start



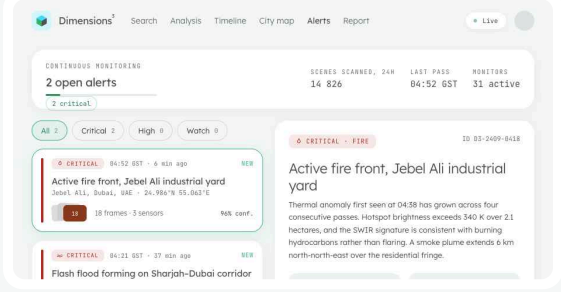
Analysis
Maps, emissions, patterns



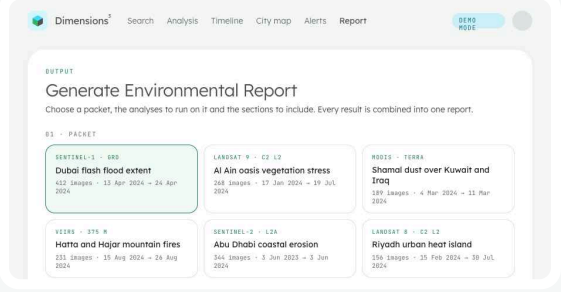
Timeline
AI digital-twin gap filling



City map
Affected streets and sites



Alerts
Live hazards for authorities



Report
One full report per packet

VALIDATION & LIMITATIONS

What supports the result, and what does not yet

VALIDATION

Cross-sensor agreement

High-agreement pixels need support from at least two sensors (0.74%). A fresh Colab run reproduces 0.574 and the MEDIUM label.

Rainfall support

Peak 53.39 mm/day, above the 50 mm threshold: score 1.0.

Reports

9 records (6 official/news + 3 public) confirm flooding in the area on the event dates.

LIMITATIONS

No ground-truth flood map yet

No independent georeferenced flood map was available. The score is heuristic; precision, recall and IoU are not yet measured.

Image timing

Optical scenes are 3–6 days after the peak, when water had drained; radar also reacts to wet soil.

Registration & false positives

A common 374 × 341 grid does not prove exact co-registration. Radar, cloud, shadow and coastline effects can create false positives. Some portal pages use sample values.

IMPACT

From days of specialist work to a clear answer anyone can read

Faster response

One pipeline turns five data sources into a flood map, a score and a summary, ready the same day.

Works through cloud

Radar keeps the service running during the storms when it matters most.

No expertise needed

Decision-makers read figures and plain sentences, not raw imagery.

Better planning

Repeated events show which roads and sites flood, guiding drainage investment.

Climate resilience

Evidence for insurance, recovery funding and climate-adaptation reporting.

Scalable

Same workflow for any UAE or GCC city, and other hazards: fire, dust, heat.

NEXT STEPS FOR INCUBATION

From one flood POC to the full Dimensions³ platform

01

Multi-satellite AI models

Analyse data from different satellites and build packets based on each user's needs.

02

Maps & emission analysis

Convert packets into maps and analysis models from spectral bands, including air emissions.

03

Digital twin

Predict the data between real images from the given observations, for a continuous timeline.

04

AI report agents

Agents that read the full packet and analysis and write clear reports.

05

Disaster prediction

Learn from patterns of past disasters to predict future ones and warn early.

06

One portal

Integrate all models into one website portal for every type of user.

Built on the POC: validate the flood pipeline first (RTC radar, co-registration, independent flood maps, precision and IoU), then extend the same workflow to every model.

Team Dimensions³ · Fatima Alkaabi · Alia Alblooshi · Wadha Alkaabi · Fatima Alnuaimi · Mariam Alneyadi

`github.com/vfatimaalkaabi/dimensions3-flood-poc`